Colloids and Surfaces A: Physicochemical and Engineering Aspects 629 (2021) 127402

Contents lists available at ScienceDirect

Colloids and Surfaces A: Physicochemical and Engineering Aspects

journal homepage: www.elsevier.com/locate/colsurfa

Lamellar crystalline networks in the gel-like phase of potassium stearate-stearic acid-water

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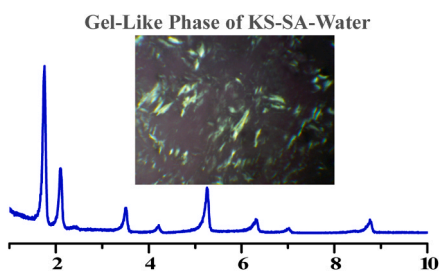
HIGHLIGHTS

- There are two isolate types of lamellar crystallites of KS and KS-SA complexes.
- Water and glycerol do not penetrate in the interlayer of lamellar crystallites.
- Network structure is formed by the crystalline fibers entrapping the solvent.
- Glycerol can promote the hydrogen bonding between the solvent and crystallites.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Fatty acid
Acid-salt complexes
Gel-like phase
Crystallite
Lamellar

ABSTRACT

The gel-like phase constructed by fatty acid compounds is a kind of fascinating soft matters but its structural characteristics have not been fully understood. This work has investigated the crystalline structures and formation mechanism of gel-like phase containing potassium stearate (KS), stearic acid (SA), and water at 80% neutralization degree. The crystalline fibers are observed in the gel-like samples of KS-SA-water on the polarized optical microscopy. X-ray diffraction measurement demonstrates that there are two isolate types of lamellar crystallites of KS and KS-SA complexes instead of the mixed crystallites in KS-SA-water samples. While KS crystallites have shorter Bragg spacing and tilted angle than the crystallites of KS-SA complexes, it is found that the solvent of water does not penetrate in the interlayer of two kinds of crystallites. The dynamic rheology measurement suggests the strong elasticity of KS-SA-water samples, which indicates the network structure formed by the lamellar crystalline fibers entrapping water. The addition of glycerol does not affect the crystalline structures of KS-SA-water samples, but significantly increase the values of storage modulus and melting enthalpy. This result suggests that the added glycerol can promote the strength of the hydrogen bonding between the solvent and crystalline fibers in the network structure of KS-SA-water samples. These findings provide new insights to the structural properties of gel-like phase of fatty acid compounds, which are importantly helpful for controlling their structure and functions.

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https://doi.org/10.1016/j.colsurfa.2021.127402
Received 11 May 2021; Received in revised form 17 August 2021; Accepted 18 August 2021
Available online 24 August 2021
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1. Introduction

Fatty acid compounds including fatty acids and their salts have attracted both industrial and academic interests [1–4]. As the typical amphiphilic molecules, the fatty acid compounds can aggregate into micelles, vesicles, liquid and solid crystalline forms. The phase behaviors of fatty acid compounds have been extensively studied [5–7], which are significantly dependent on the structure, amount, and neutralization degree of fatty acids, as well as the presence of various additives. In the temperature-composition phase diagram, it is highly characterized by the Krafft boundary which essentially gives the temperature above which the fatty acid compounds exist in the dissolved but aggregated forms such as micelles and liquid crystallites [8]. Below the Krafft temperatures, the samples with high amount of fatty acid compounds can exhibit one gel-like phase containing the chain-frozen solid crystallites and water. The gel-like fatty acid systems are a kind of fascinating soft matters which have important applications in many fields [9].

In the gel-like phase, fatty acid compounds usually crystallize into the lamellar structures below the Krafft temperatures. Besides fatty acid crystallites and fatty acid salt crystallites, many investigators have demonstrated that the crystallites of acid-salt complexes can be formed by stoichiometrically combining fatty acids and fatty acid salts through the hydrogen bonding between the carboxylic group and the carboxylate anion [10–12]. These published work focused mainly on the isolate crystallites of fatty acid, fatty acid salt, and acid-salt complexes. However, it is not known whether there are the isolate crystallites or mixed crystallites formed in the multicomponent system containing fatty acid, fatty acid salt, acid-salt complexes, and water. Furthermore, the influences of water and other solvents on the crystalline structure in fatty acid compounds have been inconsistently reported. McBain et al. observed that the crystallites of potassium laurate in a silvery white paste have nearly the same lamellar spacing as that of the anhydrous salt [13]. In the system of potassium myristate-myristic acid-water, Sagitani also found the identical spacing values of lamellar crystallites in the gel-like phase to that in the sediments separated from water [14]. However, their small angle X-ray scattering measurement demonstrated that the addition of glycerol can penetrate into the interlayer spacing of potassium myristate crystallites.

Many groups have observed the gel-like phase of fatty acid compounds, giving quite different explanations for the network structure formed by the crystallites [15,16]. Jiang et al. prepared the thermally reversible gels from the fatty acid compounds in the ionic liquid, where sodium laurate and lauric acid molecules are arranged to form the lamellar structure [17]. In the gel-like phase of sodium laurate in water and alcohol, Hao et al. explained that sodium ion together with the carboxylate headgroup of sodium laurate took part in the fabrication of the crystalline fibers [18]. Heppenstall-Butler et al. identified the crystalline structures and melting temperatures in the gel-like phase of sodium stearate-stearic acid-water [19], but the formation process of gel-like phase is not clear. On the other hand, the rheological behavior can reflect the network structure of gel-like phase of fatty acid compounds. Compared to the gel-like samples formed by the wormlike surfactant micelles and liquid crystallites [16,20], the rheological properties of gel-like samples constructed by fatty acid compounds below the Krafft temperatures are more complex and have been rarely reported in the literatures.

In order to clarify the crystalline structures and the formation mechanism of the gel-like phase in the multicomponent system of fatty acid compounds, the present work is intended to investigate the structural characteristics of a mixture of potassium stearate, stearic acid, and water at a neutralization degree of 80%. The type and structure of crystallites have been studied by polarized light microscopy, X-ray diffraction, and differential scanning calorimetry. The dynamic rheological measurements have been carried out to study the storage modulus and loss modulus of the gel-like structure of fatty acid compounds. Moreover, the effects of glycerol on the structural and rheological properties of fatty acid compounds have been identified.

2. Materials and methods

2.1. Materials

Stearic acid (SA, purity ≥ 98%) was purchased from TCI chemical Co., Ltd. Glycerol (purity ≥ 99%) was purchased from Aladdin industrial Co., Ltd. Potassium hydroxide was obtained from Shanghai Lingfeng chemical reagent Co., Ltd. The purity of potassium hydroxide was determined as 87.8%, using the calibration method with potassium hydrogen phthalate. The distilled water was used throughout the experiments.

2.2. Sample preparation of potassium stearate-stearic acid-water

The neutralization reaction between stearic acid and potassium hydroxide was carried out at 80 °C at a neutralization degree of 80%. The mass fraction of potassium hydroxide used was calculated by the following equation [21].

$$m_{KOH} = \frac{m_{SA} \times I_{SA}}{1000 \times 87.8\%} \times 80\% \tag{1}$$

where m_{KOH} and m_{SA} are the mass of potassium hydroxide and stearic acid, respectively; $I_{SA} = 196$ mg/g, which is the acid value of stearic acid. While being stirred, the mixture was maintained at 80 °C for at least 45 min to ensure the complete reaction between stearic acid and potassium hydroxide. After cooling down to 25 °C, three component samples of potassium stearate (KS)-stearic acid (SA)-water were finally obtained.

2.3. Polarized optical microscopic observation

During cooling from 80 °C to 25 °C, the microscopic texture of KS-SA-water samples was observed using a polarized optical microscope equipped with a digital camera and cross polarizers. Small amount of the samples was placed on a glass slide, which was placed on a temperature-controlled stage, so the sample temperature could be controlled under the microscope.

2.4. X-ray diffraction (XRD) measurement

The crystalline structure of KS-SA-water samples was characterized on a Bruker X-ray Powder Diffractometer. The XRD equipment was equipped with Copper Target radiation ($\lambda = 0.154$ nm) and an X-ray generator operating at 40 kV and 100 mA. The scanning area (2θ) was from 1° to 10° at a step size of 0.004°. According to Bragg's law, XRD data were displayed as the intensity (I) versus scattering vector (2θ). The Bragg spacings (d, the periodic repeat distance) in the crystalline structures, were obtained from the first diffraction peak based on the following equations.

$$d = \frac{2\pi}{q} \tag{2}$$

$$q = \frac{4\pi}{\lambda} \sin\left(\frac{\theta}{2}\right) \tag{3}$$

2.5. Rheology measurement

Rheological properties of KS-SA-water samples were measured on a Physica MCR 501 (Anton Paar) rotational rheometer fitted with parallel plate geometry (25 mm in diameter). The samples were loaded onto the plate for 10 min to allow the stress to relax and the samples to reach thermal equilibrium. The magnitudes of the storage modulus (G'), the

loss modulus (G''), and the complex viscosity (η^*) were then measured while the frequency was being varied from 1 rad/s to 62.8 rad/s. Beforehand, stain sweep tests were carried out to determine the linear viscoelastic region.

2.6. Differential scanning calorimetric (DSC) measurement

The thermodynamic properties of KS-SA-water samples were measured by a differential scanning calorimeter, Perkin-Elmer DSC7. About 5 mg sample was sealed in the standard aluminum pan, with an empty pan used as a reference. The measurement range was from 25 °C up to 80 °C at a scanning rate of 10 °C/min. While the phase transition temperatures were inferred from the peak maxima, the values of phase transition enthalpy were calculated from the area under the transition peak by the comparison with the area for a known standard (gallium).

3. Results and discussion

3.1. Preparation of potassium stearate-stearic acid-water samples

At high temperature above the Krafft temperature of fatty acids, the neutralization reaction can be easily occurred between fatty acids and salt hydroxides in the aqueous solution. When the neutralization degree is 80%, there is 80% stearic acid (SA) converted into potassium stearate (KS) together with 20% SA left. Upon cooling of the reacted mixtures from 80 °C to 25 °C, the samples of KS-SA-water can be obtained. Fig. 1 shows the light microscopic pictures of KS-SA-water samples with 5%, 10%, 20%, and 30% SA (initially added). The sample of KS-SA-water with 5% SA does not have a stable state, which is separated into the white sediment at the bottom and the clear liquid on the top in short time. Similar phenomenon was reported by Kralchevsky et al. [22], who observed the crystalline precipitates in the solution of potassium myristate. When SA amount increases from 10% to 30%, KS-SA-water samples exhibit a homogeneous gel-like appearance and no phase separation could be detected during an observation even longer than one month.

As seen in the polarized optical microscopic pictures of KS-SA-water samples with 10% and 30% SA, the crystalline fibers with obvious birefringence can be clearly seen, which can demonstrate the existence of the crystallites in the gel-like phase. The increase of SA amount from 10% to 30% leads to more bright of birefringence, which may be ascribed to the mergence of thin crystalline fibers and subsequently densely packing [15]. The formation of crystallites of fatty acids and their salts can be usually ascribed to their limited solubilities in water. Some people previously reported that fatty acid compounds can construct various crystalline structures including fibers, ribbons, platelets, and sheets, depending on the different noncovalent interactions such as electrostatic force and hydrogen bonding [8,16,23]. Terech et al. reported that crystalline fibers of fatty acid compounds have higher ability than other crystalline structures to form gel-like state [24]. For the samples of KS-SA-water with 10–30% SA, it is possibly that the interweaving of the crystalline fibers results in the formation of the gel-like state.

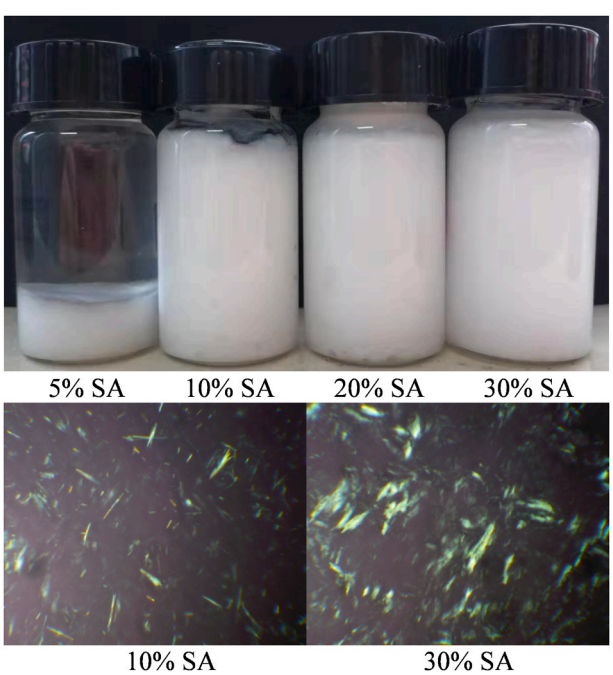

Fig. 1. Light microscopic and polarized optical microscopic pictures of KS-SA-water samples with different amounts of SA.

3.2. Crystalline structures of gel-like KS-SA-water samples

The detailed structure of crystalline fibers in the samples of KS-SA-water can be explored by using X-ray diffraction (XRD) measurement. As shown in Fig. 2, three KS-SA-water samples with 10%, 20%, and 30% SA show multiple diffraction peaks, which reflects the ordered structures of crystallites. For these samples of KS-SA-water with 80% neutralization degree, there is 80% SA converted into KS together with 20% SA left. It is well-known that fatty acids and their corresponding salts can form acid-salt complexes by the hydrogen bonding [10,12–15,25], which have different structural and physicochemical properties from respective fatty acids and salts. While acid-salt complexes have the stoichiometries of 1:1, 1:2, and 2:1 fatty acid to fatty acid salt, many investigators have confirmed the existence of only 1:1 acid-salt complexes in the system containing fatty acid and potassium salt [26,27]. Considering 80% neutralization degree of SA, besides water, there are two kinds of components in KS-SA-water samples, that is, KS and KS-SA complexes. Therefore, the multiple components in KS-SA-water samples lead to so many diffraction peaks in the XRD curves that it is hard to justify the crystalline type directly.

Eight diffraction peaks in the XRD curves of KS-SA-water samples at 80% neutralization degree can be divided into two groups in the odd and even orders, including the first to fourth orders of diffraction peaks for each group. Then, KS-SA-water samples with 10%, 20%, and 30% SA have two sets of alternate lattice spacing with the d value ratio of 1:1/2:1/3:1/4, where d is the space scattering distance calculated with

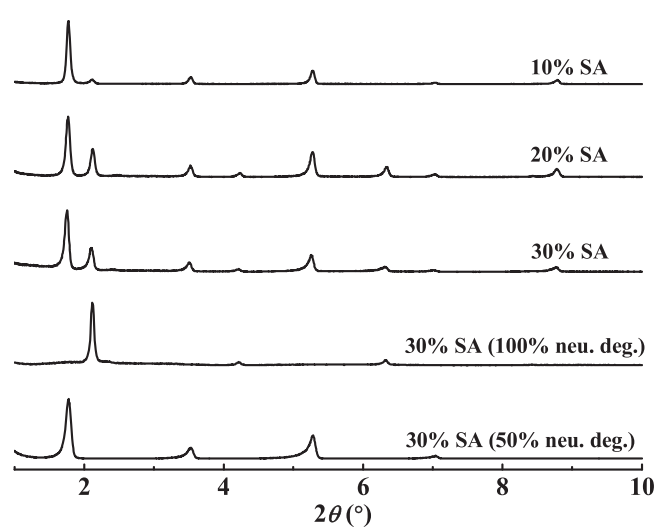

Fig. 2. XRD curves of KS-SA-water samples with 10%, 20%, and 30% SA at 80% neutralization degree. Two reference samples with 30% SA have 50% and 100% neutralization degrees, respectively.

Bragg's law. The d value ratio of 1:1/2:1/3:1/4 is the characteristic property of lamellar crystalline structure [19]. If the components form the mixed crystallites, KS-SA-water samples would give one set of diffraction peaks. Instead, in three samples two sets of diffraction peaks with the lamellar characteristics are detected, which suggests that there are two kinds of lamellar crystallines in KS-SA-water samples. Considering the components of KS and KS-SA complexes, our result can reveal that KS-SA-water samples have isolate lamellar crystallites of KS and KS-SA complexes instead of the mixed crystallites.

It is possible to further differentiate the two kinds of crystalline structures in KS-SA-water samples by comparing the XRD curves with the reference samples with the neutralization degrees of 50% and 100%, corresponding to the isolated samples of KS-SA complexes and KS, respectively. As shown in Fig. 2, both reference samples have only one set of diffraction peaks with the lamellar characteristics. Moreover, two groups of XRD peaks of KS-SA-water samples are almost respectively identical to two reference samples with the neutralization degrees of 50% and 100%. For two groups of XRD peaks, the d values of the layered structures calculated from the first diffraction peak are 50.2 Å and 41.8 Å, respectively. Consequently, the d value of 50.2 Å is attributed to the crystallites of KS-SA complexes, and KS crystallites have the d value of 41.8 Å. Despite different SA amount, the d values of lamellar crystallites are almost identical for all KS-SA-water samples, which are almost the same as those of anhydrous crystals of KS-SA complexes and KS reported by Cistola et al. [26] and Small [4], respectively. This result indicates that the solvent of water does not penetrate in the interlayer between the headgroups of KS and SA molecules in the lamellar crystallites, although KS-SA-water samples give the gel-like appearance. Norlén et al. previously demonstrated that there is essentially no water in the lamellar crystallites in palmitic system when the sample was dry or even contained 32% water [28]. Nevertheless, when the alkyl chain of fatty acid compounds is long enough, their crystalline phases do not swell in water. Thus, the high hydrophobicities of KS and SA molecules should be the reason for the fact that the presence of water in KS-SA-water samples does not affect the Bragg spacings both in KS crystallites and in the crystallites of KS-SA complexes.

In the lamellar crystallites of KS and KS-SA complexes, the molecules of KA and SA align themselves along the alkyl chain axis and pack into bilayers. For the fatty acid compounds with a fully stretched linear chain, the chain length can be estimated by the following equation proposed by Tanford [29].

$$l(\text{Å}) = 1.5 + 1.265n \tag{4}$$

where 1.5 Å is the van der Waals radius of the terminal methyl group ($-CH_3$), 1.265 Å is the distance between two adjacent carbon atoms along a rigid alkyl chain axis, and n is the number of the carbon atoms in the inner methyl groups ($-CH_2-$). Moreover, the size of a carboxylic group is 4.2 Å. Thus, the length of KS including alkyl chain and headgroup is approximately 25.9 Å. Assuming the alkyl chains are packed perpendicularly to the plane of the bilayer, the d-spacing of the bilayer structure is approximately 51.9 Å.

However, our XRD data reveal that both KS crystallites and the crystallites of KS-SA complexes give shorter d values, which can indicate that the alkyl chains of KS and SA are tilted with respect to the basal planes [30]. Here, KS crystallites are calculated to have a tilted bilayer structure with a repeat distance of 41.8 Å and a tilted angle of 53.6°, which are consistent with the reported values for KS crystallites in the literatures [4,31]. On the other hand, the d value of 50.2 Å in the crystallites of KS-SA complexes is matched well with the data calculated for the Bragg spacings of the crystallites of acid-salt complexes from the following equation proposed by Small [4].

$$d = 2.54n + 4.7\text{Å} \tag{5}$$

where n is the number of carbons in the alkyl chain. Comparing to the d value of 51.9 Å for the perpendicularly packing of the alkyl chains to the plane of the bilayer, the crystallites of KS-SA complexes have a tilted bilayer structure with a repeat distance of 50.2 Å and a tilted angle of 75.3°. The d value of 50.2 Å in the crystallites of KS-SA complexes is longer than the d value of 41.8 Å in KS crystallites, which is consistent with other fatty acid systems [26].

3.3. Rheological properties of gel-like KS-SA-water samples

Rheological behaviors have important values to understand the gel-like structure of KS-SA-water samples. All the gel-like KS-SA-water samples show similar rheological properties. Fig. 3 shows the typical result of small deformation oscillatory measurement of KS-SA-water sample with 30% SA. The complex viscosity of the sample is observed to decrease gradually with the oscillation frequency, which indicates the general shear thinning behavior. Remarkably, the storage modulus (G') is found to be more than 2 times the loss modulus (G'') at all frequencies measured. Liquid crystallites of fatty acids and their salts usually have the viscoelastic behavior, that is, G'' exceeds G' at lower frequencies; at higher frequencies G' has bigger values than G'' [32,33]. However, the significantly higher G' than G'' and the absence of the intersection may demonstrate the predominantly elastic property of gel-like KS-SA-water samples. This result is consistent with the observation in sodium stearate system reported by Davis et al. [30] The crystalline fibers may entangle themselves with one another to form the highly interconnected network structure, accounting for the strong elasticity of KS-SA-water samples. On the other hand, the values of G', an important parameter to characterize the gel strength, become higher as SA amount increases from 10% to 30%. Because the elasticity is attributed to the strong network of

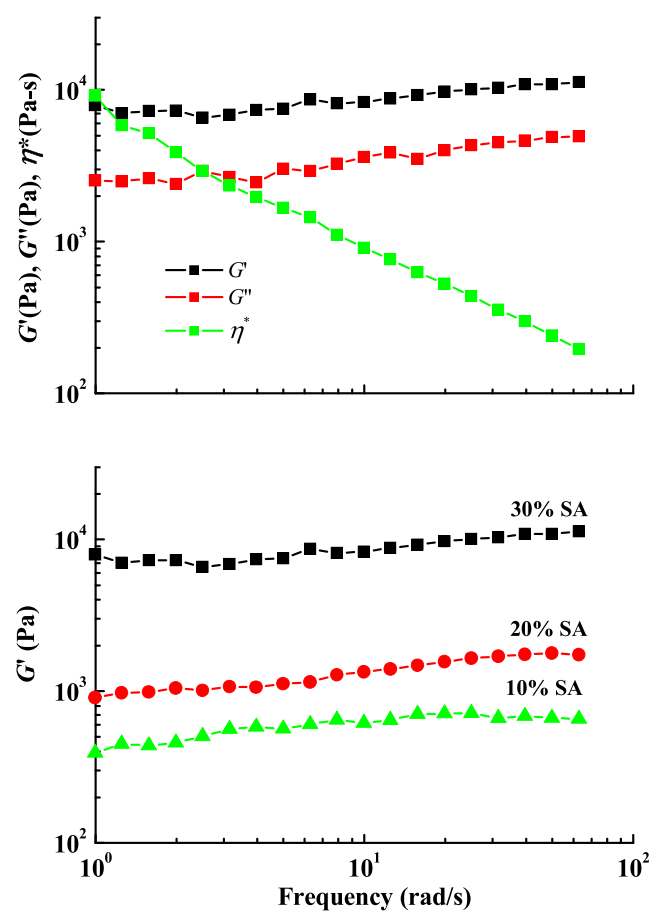

Fig. 3. (a) The storage modulus G', the loss modulus G'', and the complex viscosity η^* versus angular frequency for KS-SA-water sample with 30% SA. (b) The storage modulus G' versus angular frequency for KS-SA-water samples with 10%, 20%, and 30% SA.

crystalline fibers, the variation of G' values suggests that three-dimensional networks become much denser with an increase of SA concentration. In addition, G' values increase slightly with increasing frequency. In the liquid crystallite systems, orientational ordering over the sample-size length scales can be obtained during the uniaxial deformation in the external field such as a shear flow [34,35]. The result of the enhanced G' values may be a signature of highly ordered packing of crystallites upon increasing frequency.

3.4. Heating behavior of Gel-like KS-SA-water samples

The thermal behavior of KS-SA-water samples has been measured as a function of temperature using differential scanning calorimetry (DSC). Opposite to the cooling process during the sample preparation, heating may cause KS-SA-water samples to change from the gel-like state to the liquid phase. Fig. 4 shows the DSC heating curves of KS-SA-water samples with 10%, 20%, and 30% SA at 80% neutralization degree, together with two reference samples with 30% SA at 50% and 100% neutralization degrees. Three KS-SA-water samples with 10%, 20%, and 30% SA at 80% neutralization degree have two endothermic peaks in their DSC curves, which is consistent with the observation for the samples containing partially neutralized fatty acid reported by Heppenstall-Butler et al. [19] The two reference samples with the neutralization degrees of 50% and 100%, corresponding to the isolated samples of KS-SA complexes and KS respectively, have only one endothermic peaks at 67.8 °C and 57.1 °C. This result is consistent with the observation by Kung et al. [31], regarding that the crystallites of acid-soap complexes have higher melting temperature than the soap crystallites. For the samples of KS-SA-water, the first and second endothermic peaks can be ascribed to the phase transitions of KS crystallites and the crystallites of KS-SA complexes, respectively. Upon heating, the endothermic peaks of KS-SA-water samples may be resulted from a complex process that includes the break-up of the crystalline network and melting of the solid crystallites into the liquid form. The phase transition temperatures and enthalpy changes of KS-SA-water samples have been inferred from the peak maxima and the peak area, as listed in Table 1.

In the DSC curves of KS-SA-water samples, it is clearly seen that the first endothermic peak moves toward higher transition temperature when SA amount increases from 10% to 30%, and conversely, the second endothermic peak shifts to lower transition temperature. From the almost identical XRD peaks for KS-SA-water samples with 10%, 20%, and 30% SA, we know that KS crystallites and the crystallites of KS-SA complexes in three samples do not have marked changes with increasing SA amount. Thus, the mergence tendency of two transition peaks for KS-SA-water samples may indicate the strong entanglement of two kinds of crystallites at high amount of SA. Because of the more closely packing of crystallites, the enthalpy changes of KS-SA-water samples also show the increased values at high SA amount. During the phase transition induced by heating, the substantial rearrangements of the headgroup structure in the crystallites of KS and KS-SA complexes may take place, which can be supported by the enthalpy changes in the hydrogen bonding range.

Table 1
Phase transition temperatures and corresponding enthalpy changes of KS-SA-water samples.

KS-SA-water samples	T_1 (°C)	ΔH_1 (J/g)	T_2 (°C)	ΔH_2 (J/g)
10% SA	51.5	2.7	64.7	3.5
20% SA	52.5	7.4	63.1	3.9
30% SA	54.2	8.3	60.2	7.1
30% SA + Glycerol	50.1	28.7	71.5	20.5

3.5. Effect of glycerol addition

Glycerol is one polyhydric alcohol commonly added in the products of fatty acids and fatty acid salts [36]. Sagitani et al. found that the addition of glycerol can expand the surfactant concentration range for the lamellar gel phase of myristic acid and potassium myristate [14]. Stuart et al. observed that changing solvent from water to glycerol can induce the morphological change of crystallites of fatty acid compounds [37]. In order to investigate the effect of glycerol on the structural properties of KS-SA-water samples, 30% glycerol was added to KS-SA-water sample with 30% SA, which was prepared by the same

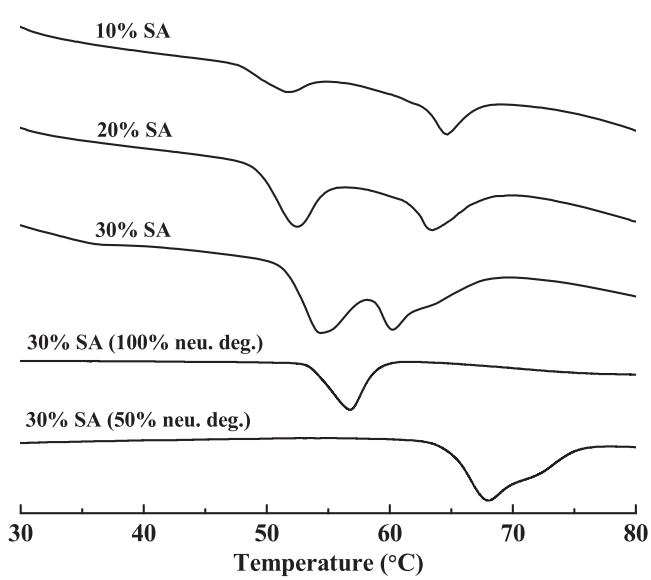

Fig. 4. DSC heating curves of KS-SA-water samples with 10%, 20%, and 30% SA at 80% neutralization degree. Two reference samples with 30% SA have 50% and 100% neutralization degrees, respectively.

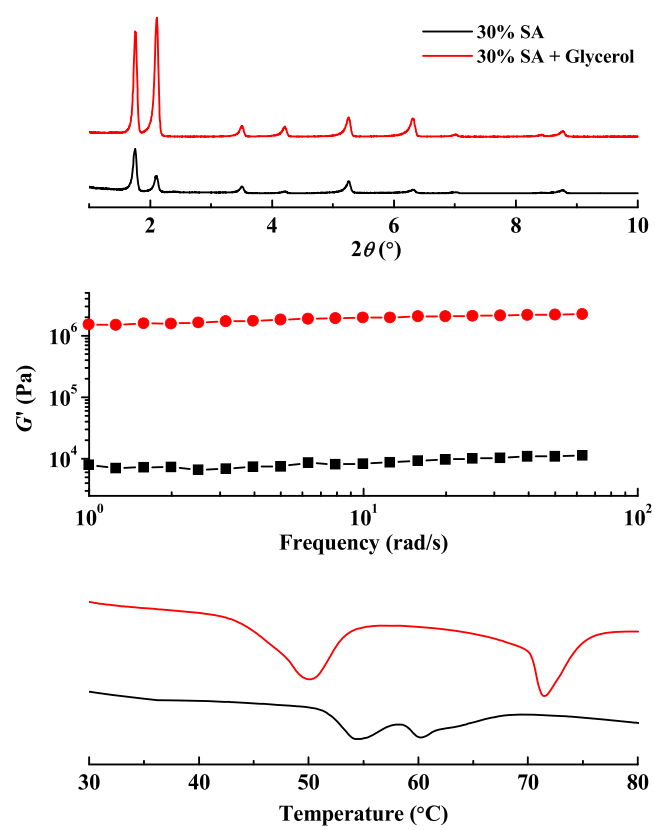

Fig. 5. Effect of glycerol on the XRD, rheology, and DSC heating curves of the sample of KS-SA-water with 30% SA.

procedure without glycerol. As shown in Fig. 5, the XRD curve of the sample with glycerol has the multiple diffraction peaks, giving almost same diffraction positions as the sample without glycerol. This result indicates that neither the crystalline type nor the spacings of two lamellar crystallites are affected by the addition of glycerol. However, it is noted that all diffraction peaks for the sample with glycerol have more enhanced diffraction intensities than the sample without glycerol, especially for the second group of diffraction peaks corresponding to KS crystallites. At high temperatures before the formations of lamellar crystallites, the headgroups of KS molecules and KS-SA complexes have the weakened electrostatic repulsions in the mixed solvent of glycerol and water compared to pure water. Thus, KS molecules and KS-SA complexes have big tendency to form longer lamellar crystalline fibers upon cooling, which may be the reason for the enhanced diffraction peaks in the XRD curve of the sample containing glycerol. Because KS molecules carry more charges than KS-SA complexes, the addition of glycerol gives more reduction in the electrostatic repulsion between the headgroups of KS molecules. Therefore, the growth of KS crystallites is more significantly promoted by the added glycerol than the crystallites of KS-SA complexes.

The storage modulus curves of KS-SA-water samples with and without glycerol have been also compared in Fig. 5. The KS-SA-water sample with glycerol has markedly higher G' values compared to the sample without glycerol. Although the mixed solvent of glycerol and water cannot penetrate into the crystallites, the multiple hydrogen bonds may be formed between glycerol and the adjacent KS molecules or KS-SA complexes in the surfaces of crystallites. Therefore, when glycerol is added, the hydrogen bondings of crystallites with solvent can strengthen the aggregation of longer crystalline fibers into more ordered structure as a gel-like state. Meanwhile, the markedly increasing of G' values with the increase in the frequency is mainly resulted from the high elasticity of crystallites of KA-SA complexes, which is supported by the rise of the second endothermic peak in the DSC curve of KS-SA-water sample with glycerol.

As shown in Fig. 5, the addition of glycerol results in more separation of two endothermic peaks for KS-SA-water sample containing glycerol. Compared to the sample without glycerol, the first endothermic peak of KS-SA-water sample with glycerol moves to lower temperature, whereas the second endothermic peak moves to higher temperature. The phase transition temperatures and enthalpy changes of KS-SA-water sample with glycerol are listed in Table 1. As discussed above, the addition of glycerol may promote the growth of crystalline fibers formed by KS-SA complexes due to the suppression of electrostatic repulsions. This may be the reason for the increased phase transition temperature corresponding to the crystallites of KS-SA complexes. Although the addition of glycerol has more pronounced effect on the growth of KS crystallites than the crystallites of KS-SA complexes, the added glycerol may give more chances for the hydrogen bonding between KS crystallites and solvent. Because the hydrogen bonding may be destroyed by heating, the addition of glycerol may cause the networks of KS crystallites to break out and change into liquid type of crystallites at lower temperature. For two endothermic peaks, the enthalpy values of KS-SA-water sample with glycerol is 2.9–3.5 times those of the sample without glycerol. This observation is consistent with the result from Davis et al. [30], who reported an increase in the dissolution enthalpy for long-chain fatty acid compounds upon changing from water to the mixed solvent of water and propylene glycerol.

3.6. Formation mechanism of the network structure of KS-SA-water samples

While SA can act both as a proton donor and as an acceptor, the carboxylate anion of KS is a stronger proton acceptor. In KS-SA-water samples at 80% neutralization degree, KS-SA complexes are formed through the intermolecular hydrogen bonding between KS and SA at a 1:1 ratio [26,31]. Because of the high insolubility resulted from the long alkyl chain containing 18 carbons, KS and KS-SA complexes appear in two isolate types of lamellar crystallites, as demonstrated by XRD results. The crystallites of KS have the lamellar structure, which consists of the bilayers with the hydrocarbons separated by the sheets of polar constituents [38]. The small potassium ions bind to and immobilize the carboxylate headgroups, so the headgroup of carboxylate ions and the potassium ions arrange periodically on the surface of the cylindrical fibers. However, the lamellar crystallites of KS-SA complexes maintain some carboxylic acid-like character and some carboxylate anion-like character, which are quantitatively different from either the parent fatty acid or potassium carboxylate [12]. Carboxylate and acid-like pairs of KS-SA complexes are arranged in pairwise fashion to form a bilayer structure, accommodating a “short” hydrogen bond, and potassium ions are shared among laterally adjacent carboxylate anions to create a pseudo-six-member ring, which can add to the crystal stability.

XRD measurements further reveal that KS crystallites have a tilted bilayer structure with a repeat distance of 41.8 Å and a tilted angle of 53.6°, while the crystallites of KS-SA complexes have a repeat distance of 50.2 Å and a tilted angle of 75.3°. Although the alkyl chain types are matched in KS and KS-SA complexes, KS molecules have one negative charge per headgroup, whereas KS-SA complexes have one negative charge per every two headgroups. The number of headgroup charges in KS and KS-SA complexes determines the bilayer thickness and the tilt of the hydrophobic region in two types of crystalline fibers formed by KS and KS-SA complexes [30]. The hydrated KS experiences a greater lateral charge repulsion between polar groups in the plane of the bilayer, giving rise to greater surface areas per polar group in crystalline phase. Since KS-SA complexes experiences less charge repulsion, the crystalline fibers of KS-SA complexes that are formed can accommodate smaller surface areas. It was reported that the solvent can penetrate in the interlayer between the headgroups of surfactants of liquid crystallites and even solid crystallites made by surfactants with relatively short alkyl chain [7,39]. Nevertheless, our XRD measurements reveal that the solvents of water and glycerol does not affect the thickness of bilayer (the total thickness of hydrophobic domain and hydrophilic headgroups) not only for KS crystallites but also for the crystallines of KS-SA complexes. The crystalline phases in the gel-like samples of KS-SA-water strongly indicate the high hydrophobicity of the long alkyl chain in KS and KS-SA complexes.

Many investigators have reported that the fibers of long-chain fatty acid compounds can form thermally reversible gels in water and polar organic solvents [18,23,30,37,40]. The fibers of long-chain fatty acid compounds are considered to be originated from the lamellar crystallites in the growth of fast infinite length compared to the growth in fiber width and fiber thickness. Two kinds of crystalline fibers of KS and KS-SA complexes may entangle themselves with one another to form the network to jellify the solvent of water [41], as indicated by the gel-like appearance of KS-SA-water samples. In the gel-like samples of sodium methyl stearoyl taurate-behenyl alcohol-water system, Watanabe et al. found that there are two kinds of water, including the water solubilized in the interlayer and trapped in the matrix of α-crystallites [42]. Although the solvent of water is not almost penetrated in the lamellar structures of the crystalline fibers of KS and KS-SA complexes, water may form hydrogen bonds with carboxylic and carboxylate groups of KS and SA molecules in the surfaces of crystallites [26]. The solvent of water is entrapped in the interspaces of the three-dimensional network of the crystalline fibers of KS and KS-SA complexes by surface tension and capillary forces [18,23]. The strong elasticity of KS-SA-water samples is resulted from the network structure formed by the crystallites. The increase of SA amount can not only produce more crystallites, but also cause the crystalline fibers to be packed more densely, even interconnected to form some junction dots [15]. Therefore, the values of G' and the melting enthalpy give the increased changes with the increasing SA amount [17]. On the other hand, with three hydroxyl groups, glycerol can increase the hydrogen bonding aggregation of crystalline fibers with solvent molecules, which is helpful in forming the network

structure in the gel-like phase.

4. Conclusions

Differently from the extensively studied phase behaviors of fatty acid compounds, this work has yielded comprehensive information on the crystalline structures and the formation mechanism of the gel-like phase of KS-SA-water at a neutralization degree of 80%. Qualitatively, all experimental data prove that the gel-like KS-SA-water samples have the network structure formed by the lamellar crystalline fibers entrapping water. Two isolate types of lamellar crystallites of KS and KS-SA complexes instead of the mixed crystallites are demonstrated. Although the solvents of water and glycerol do not penetrate in the crystallites, the addition of glycerol contributes a great deal to strengthen the gel network of KS-SA-water samples through the hydrogen bonds with KS molecules or KS-SA complexes in the surfaces of crystallites. Because the gel-like fatty acid systems are fascinating soft matters, the new insight to the structural properties of KS-SA-water samples is helpful for the control of the structure and functions of gel-like phase of fatty acid compounds used in various fields.

CRediT authorship contribution statement

Heliang Du: Methodology, Investigation, Data curation, Formal analysis, Writing – original draft. **Xiaoyong Wang:** Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by NIVEA (Shanghai) Co. Ltd., China. Dr. Wang would like to acknowledge the National Natural Science Foundation of China (Grant 21573071).

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